

PROJECT PROPOSAL (Proposed Solution)

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Project Title: Diabetic Retinopathy Classification Web Application

Maximum Marks: 3 Marks

1. Introduction

Diabetic Retinopathy (DR) is a serious eye disease caused by complications of diabetes. If not detected early, it can lead to vision loss or blindness. Early diagnosis and classification of DR are essential for timely treatment.

This project proposes a Deep Learning based web application that classifies retinal images into different stages of Diabetic Retinopathy using a Convolutional Neural Network (CNN) model.

2. Problem Overview

Manual screening of retinal images is time-consuming and requires expert ophthalmologists. In many rural or underdeveloped areas, access to specialists is limited.

There is a need for an automated system that can:

- Analyze retinal images
 - Detect presence of DR
 - Classify severity level
 - Provide fast and reliable results
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3. Proposed Solution

The proposed solution is a Flask-based web application integrated with a trained CNN model.

The system will:

1. Allow users to upload retinal images
 2. Preprocess the image
 3. Pass the image to trained CNN model
 4. Predict DR severity
 5. Display result on screen
 6. Save result into database
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4. System Architecture

The system will consist of:

- User Interface (HTML/CSS)
- Flask Backend
- Deep Learning Model (.h5 file)
- SQLite Database

Workflow:

User → Upload Image → Flask → CNN Model → Prediction → Database → Display Result

5. Advantages of Proposed System

- Faster diagnosis
 - Reduces manual effort
 - Improves screening efficiency
 - Supports medical professionals
 - Scalable and cost-effective
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6. Conclusion

This project will provide an intelligent and automated solution for Diabetic Retinopathy detection. By integrating Deep Learning with a web application, the system will help in early detection and efficient medical support.